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DISCRETE MATHEMATICS & APPLICATIONS

Based on Kenneth H. Rosen, 8th Edition

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Topic

1. Propositional Logic and Proofs.
2. Propositional Equivalence.
3. Predicates and Quantifier.
4. Graph.
5. Tree.
6. Number Theory.
7. Cryptography.
8. Set.
9. Function.

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Introduction to
Discrete Mathematics

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What is Discrete Mathematics ?

Definition

Discrete Mathematics is the study of **discrete objects**.

Discrete VS Continuous

Discrete	Continuous
1. Can’t be broken down into fraction or decimals.	1. Can be broken down into fraction or decimals.
2. Number of students, purchasing tickets, message etc.	2. Time, temperature, weight, distance etc.
3. This elements are countable.	3. This elements are measurable.

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Why do you need to study discrete Mathematics?

1. It is very good tool for improving problem solving capability.

Mathematical Logic

Propositions, truth tables, and rules of logical inference.

Graph Theory

Vertices and edges representing networks, social connections, internet routing, and data structures.

Number Theory

Find GCD with Euclidean Algorithm, prime numbers, modular arithmetic—the absolute core of modern data security and **Cryptography**.

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WHAT IS A PROPOSITION?

- A **Proposition** is a declarative sentence that is either **True** or **False**.
- Crucial Constraint: It cannot be both True and False at the same time.
- Examples: "2 + 2 = 4" (True), "The moon is made of cheese" (False).
- Non-examples: "Open the door!" (Command), "What time is it?" (Question).

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THE NEGATION

Definition

Let p be a proposition. The **Negation** of denoted by **¬ P** , is the statement:

"It is not the case that P"

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NEGATION EXAMPLE & TABLE

P: Today is Monday.

¬ **P:** Today is **not** Monday.

OR, ¬ **P:** It is not the case that today is Monday.

P	¬ P
T	F
F	T

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Conjunction (AND)

Joining propositions with
"and"

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LOGICAL AND

Definition

Let p and q be proposition. The conjunction of p and q, is the proposition "p and q". The conjunction **p ∧ q** is true when both p and q are true and is false otherwise.

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CONJUNCTION: EXAMPLE & TABLE

p : Today is Monday.

q : It is raining.

$p \land q$ **:** Today is Monday **and** it is raining.

p	q	p ∧ q
T	T	T
T	F	F
F	T	F
F	F	F

Note: All inputs must be True for a True output.

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Disjunction (OR)

Joining propositions with "or"

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LOGICAL OR

Definition

Let p and q be proposition. The disjunction of p and q, is the proposition "p or q". The disjunction **p V q** is false when both p and q are false and is true otherwise.

$$p \vee q$$

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DISJUNCTION : EXAMPLE & TABLE

p : Students who have taken Calculus.

q **:** Computer Science can enroll.

p V q : Students who have taken Calculus **or** Computer Science can enroll

p	q	p V q
T	T	T
T	F	T
F	T	T
F	F	F

Note: If at least one input is True, the output is True.

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Exclusive OR (XOR)

Definition

Let p and q be proposition. The exclusive or of p and q, denotes by "p $\oplus$ q" , is the proposition that is true when exactly one of p and q is true and is false otherwise.

$p \oplus q$

p	q	p $\oplus$ q
T	T	F
T	F	T
F	T	T
F	F	F

Example:

"Students who have taken calculus or computer science, but not both, can enroll the clsss."

Conditional Statement / Implication

Definition

Let p and q be proposition. The conditional statement p → q is the proposition if p then q. The conditional statement p → q is false when p is true when p is true and q is false and true otherwise. In conditional statement p → q, p is called the hypothesis and q is called the conclusion.

$$p \to q$$

p: Hypothesis (Antecedent)

q: Conclusion (Consequent)

p	q	p → q
T	T	T
T	F	F
F	T	T
F	F	T

Note: TF = F , Otherwise True.

Example:

"If I will elected, then I will lower taxes."

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Representing p → q

1. If p, then q.
2. If p, q.
3. p is sufficient for q.
4. q if p.
5. q when p.
6. A necessary condition for p is q.
7. q unless ¬p.
8. p implies q.
9. p only if q.
10. A sufficient condition for q is p.
11. q whenever p.
12. q is necessary for p.
13. q follows from p.

Converse, Inverse, Contrapositive

From conditional statement :

Conditional Statement

$p \rightarrow q$

Equivalent

Contrapositive

$\neg q \rightarrow \neg p$

Converse

$q \rightarrow p$

Equivalent

Inverse

$\neg p \rightarrow \neg q$

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Example:

Conditional Statement: $p \to q$

If it is raining, then the home team wins.

Converse: $q \to p$

If the home team wins, then it is raining.

Contrapositive: $\neg q \to \neg p$

If the home team does not win, then it is not raining.

Inverse: $\neg p \to \neg q$

If it is not raining, then the home team does not win.

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Equivalence Truth Table

p	q	¬p	¬q	p → q Conditional Statement	¬p → ¬q Inverse	¬q → ¬p Contrapositive	q → p Converse
T	T	F	F	T	T	T	T
T	F	F	T	F	T	F	T
F	T	T	F	T	F	T	F
F	F	T	T	T	T	T	T

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Biconditional Statement

Definition

Let p and q be proposition. The biconditional statement $p \leftrightarrow q$ is the proposition p if and only if q. The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values and is false otherwise.

$$\boxed{p \leftrightarrow q}$$

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example:

"You can take the flight if and only if you buy a ticket."

$$\boxed{p \leftrightarrow q} \Rightarrow (p \to q) \land (q \to p)$$

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Different ways :

I. p is necessary and sufficient for q.

II. if p then q, and conversely.

III. P iff q.

Complex Truth Table

Construct table for: $(p \lor \neg q) \rightarrow (p \land q)$

p	q	¬q	p ∨ ¬q	p ∧ q	(p ∨ ¬q) → (p ∨ ¬q)
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F

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Precedence of logical operators

Rank	Symbol	Name
1	¬	Negation
2	∧	Conjunction (AND)
3	∨	Disjunction (OR)
4	→	Implication
5	↔	Biconditional

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Logic Examples: Set 1

p: It is below freezing.
q: It is snowing.

a. It is below freezing and snowing. $p \land q$

b. It is below freezing but not snowing. $p \land \neg q$

c. It is not below freezing and it is not snowing.
$\neg p \land \neg q$

d. It is either below freezing or it is snowing, but it is not snowing if it is below freezing.
$(p \lor \neg q) \rightarrow (p \land q)$

e. That it is below freezing is necessary and sufficient for it to be snowing. $p \leftrightarrow q$

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Logic Examples: Set 2

p: Yor drive over 65 miles per hour.
q: You get a speeding ticket.

a. You do not drive over 65 miles per hour. $\neg p$

b. You drive over 65 miles per hour, but you do not get a speeding ticket. $p \land \neg q$

c. Driving over 65 miles per hour is sufficient for getting a speeding ticket. $p \rightarrow q$

d. You get a speeding ticket, but you do not drive over 65 miles per hour. $q \land \neg p$

e. Whenever you get a speeding ticket, you are driving over 65 miles per hour. $q \leftrightarrow p$

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Complex Translation Example

"You can access the internet from campus **only if** you are a computer science major **or** you are not a freshman."

p: You can access the internet from campus.

q: You are a computer science major.

r: You are a freshman.

$$p \rightarrow (q \lor \neg r)$$

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Propositional Equivalences

(Tautology ,Contradiction, Contingency)

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Definition

1. **Tautology:**

A compound proposition that is always true , no matter what the truth value of the proposition that occur I it, is called a tautology.

2. **Contradiction:**

A compound proposition that is always false is called a Contradiction.

3. **Contingency:**

A compound proposition that is neither a tautology or nor a contradiction is called a Contradiction.

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Solution

p	q	⌐ p	p → q	⌐p V q
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Example

- Show that p → q and ⌐ p V q are logically equivalent.

p → q ≡ ⌐ p V q

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Logical equivalence

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1. Identity laws:

$p \land T \equiv p$

$p \lor F \equiv p$

2. Domination laws:

$p \lor T \equiv T$

$p \land F \equiv F$

3. Double negation laws:

$\neg(\neg P) \equiv p$

4. Commutative laws:

$p \lor q \equiv q \lor p$

$p \land q \equiv q \land p$

5. Associative laws:

$(p \lor q) \lor r \equiv p \lor (q \lor r)$

$(p \land q) \land r \equiv p \land (q \land r)$

6. Distributive laws:

$p \lor (q \lor r) \equiv (p \lor q) \land (p \lor r)$

$p \land (q \lor r) \equiv (p \land q) \lor (p \land r)$

7. De morgan's laws:

$\neg(p \lor q) \equiv \neg p \land \neg q$

$\neg(p \land q) \equiv \neg p \lor \neg q$

8. Absorption laws:

$p \lor (p \land q) \equiv p$

$p \land (p \lor q) \equiv p$

$p \rightarrow q \equiv \neg p \lor q$

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Predicates and Quantifier

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Predicates

$x > 3$

$x + y = 2$

$x + y > z$

Every student in this class has studied calculus.

$\underbrace{\text{X}}_{\text{Subject}}$ $\underbrace{\text{is grater then 3.}}_{\text{Predicates}}$

P(x) = Propositional function

P(x)= x is grater then 3.

Let Q (x , y) denoted the statement $x = y + 3$ what are the truth value of the propositions Q (1,2) and Q (3,0) ?

Solution:

$Q(x, y) \rightarrow x = y+3$

$Q(1,2) \rightarrow 1 = 2+3 = F$

$Q(3,0) \rightarrow 3 = 0+3 = T$

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Quantifier

1. Universal quantifier:

$\forall x$ P (x) = For all x p (x).

2. Existential quantifier:

$\exists x$ P(x) = There exist an x such that p(x).

Example 1:

Let p(x) be the statement $x + 1 > x$. What is the truth value of $\forall x$ p(x), where the domain consist of all real numbers.

$\forall x$ p(x) = True

P(x) = $x+1>x$

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Example 2:

(a). Let Q(x) be the statement $x < 2$. What is the truth value of $\forall x$ Q(x), where the domain consist of all real numbers.

$\forall x$ Q(x) = False

(b). Let Q(x) denoted the statement "$x = x + 1$". What is the truth value of $\exists x$ Q(x); where the domain consists of all real numbers?

Q(x) → $x = x + 1$

$\exists x$ Q(x) = False

Example

1. P(x) = x has taken a course is calculus domain students of your class.

$\forall x$ P(x)

2. For all students x in your class has taken a course in calculus. Every student in your class has taken a course in calculus.

$\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$

Or, $\neg\, \exists x\, P(x) \equiv \forall x\, \neg P(x)$

It is not the case that every student in your class has taken a course in calculus.

Or, There is a student in your class who has not taken a course in calculus.

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Graph

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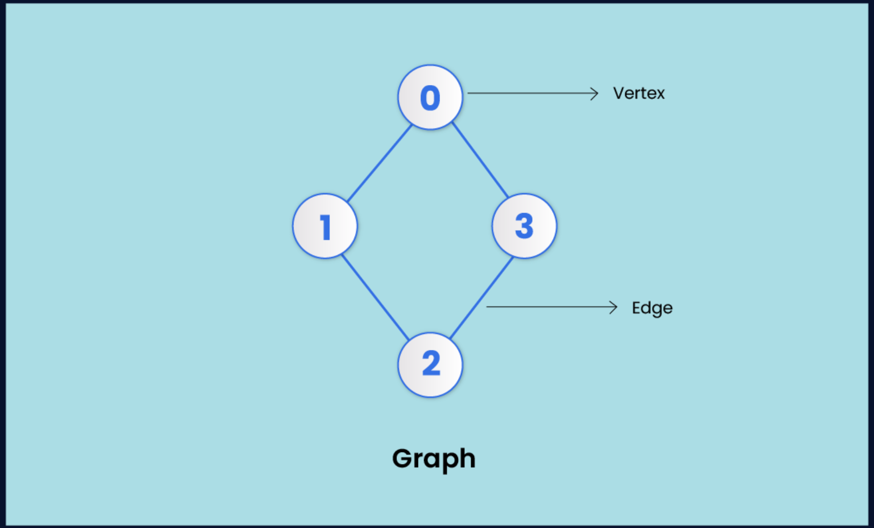
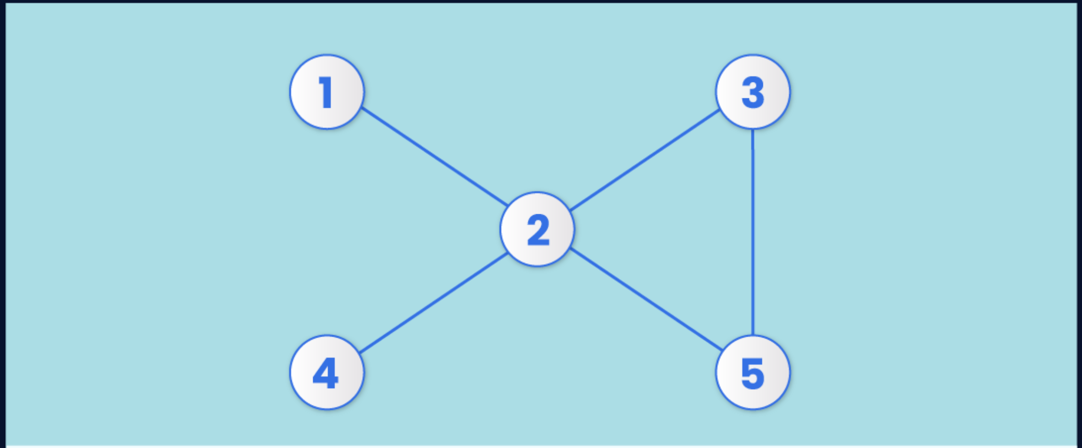
Definition

A graph G = (V,E) consists of V a nonempty set of vertices (or nodes) and E, a of edges.



Finite graph:

If the number of vertices and edges is limited.



FINITE GRAPH

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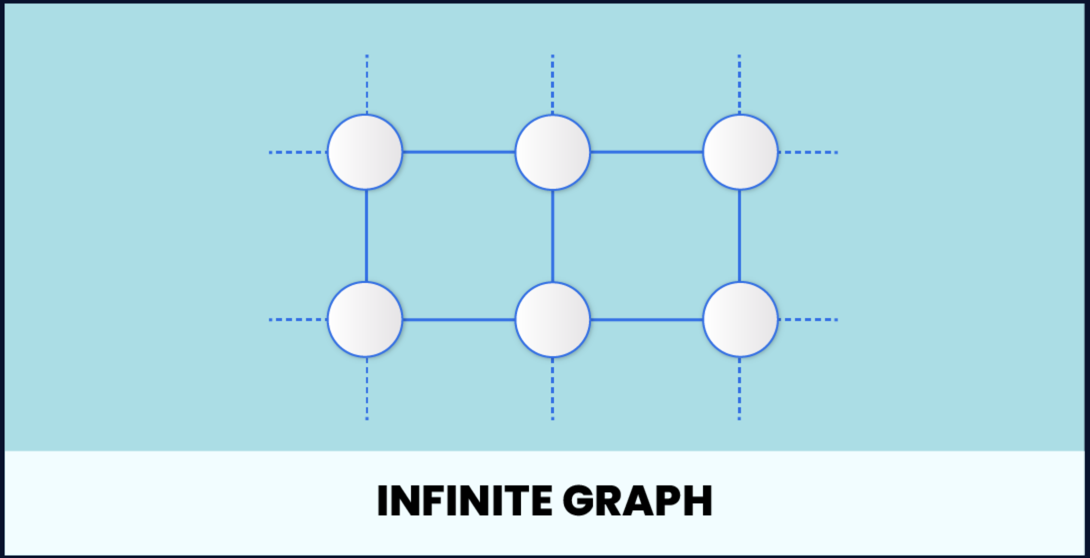
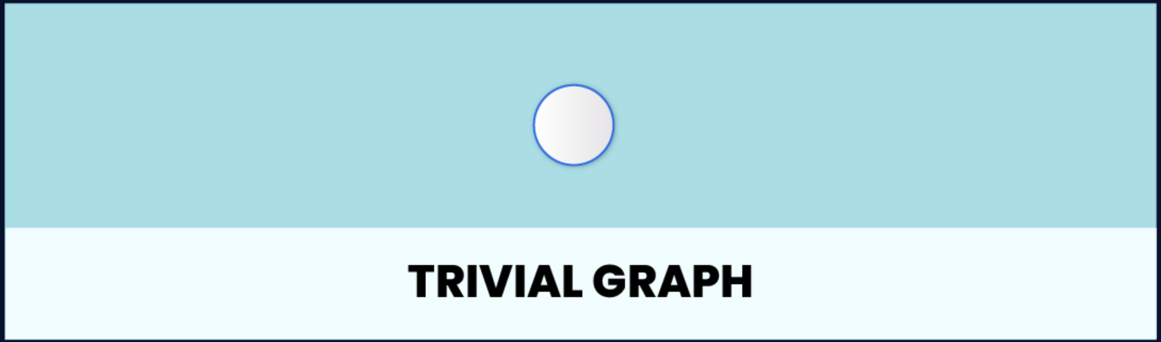
Infinite graph:

If the number of and edges is unlimited.



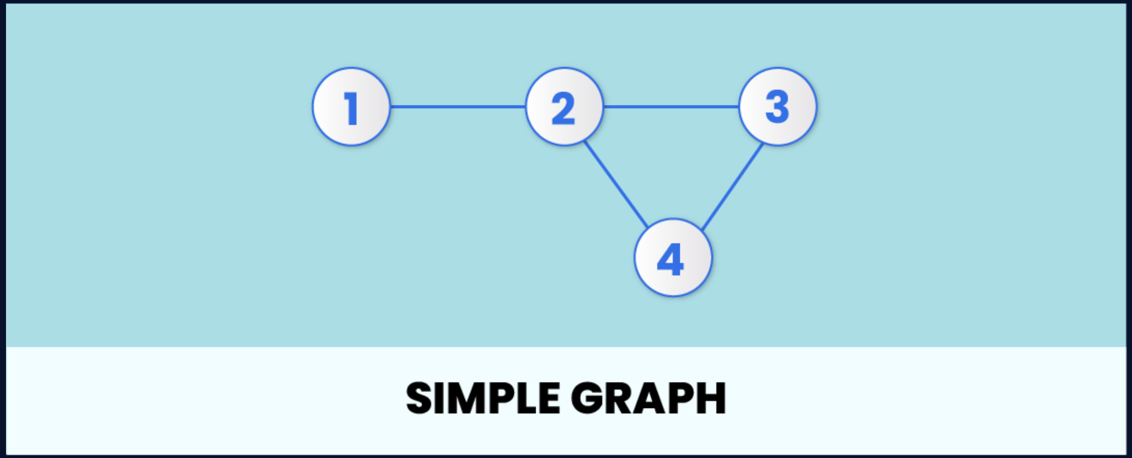
Trivial graph:

If it contains any a single vertex and no edges.



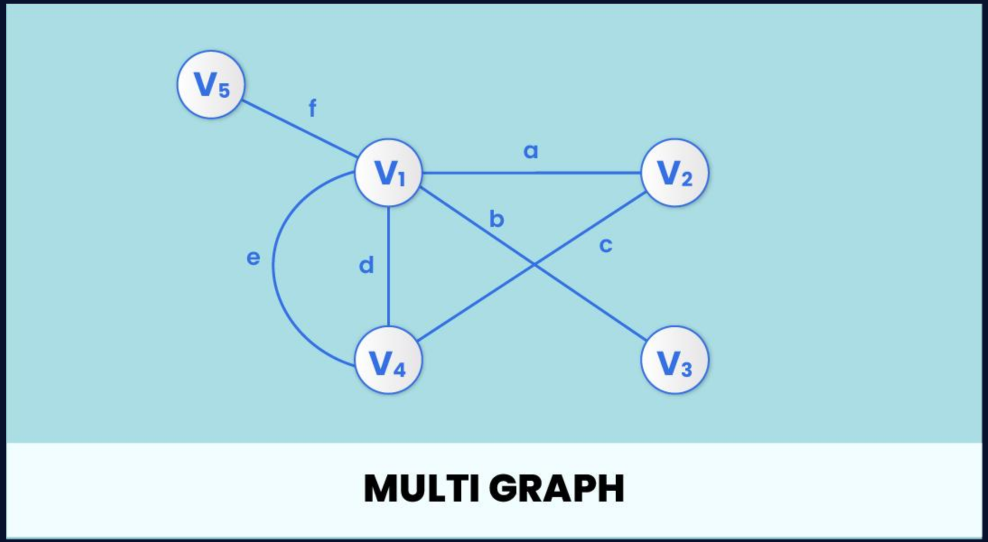
Simple graph:

If each pair of vertices has only one edge.



Multi graph:

If there are numerous edges between a pair of vertices.



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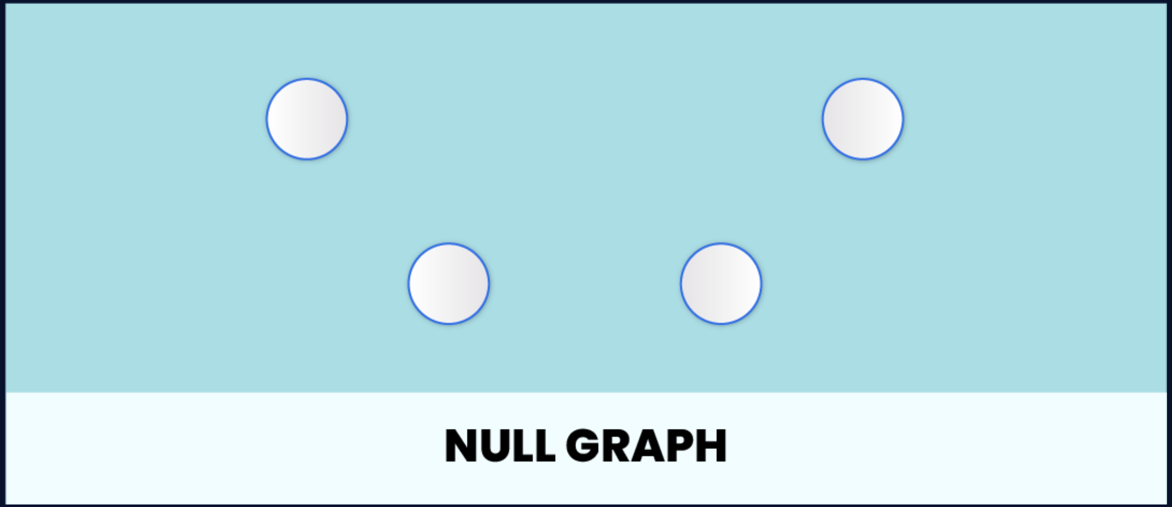
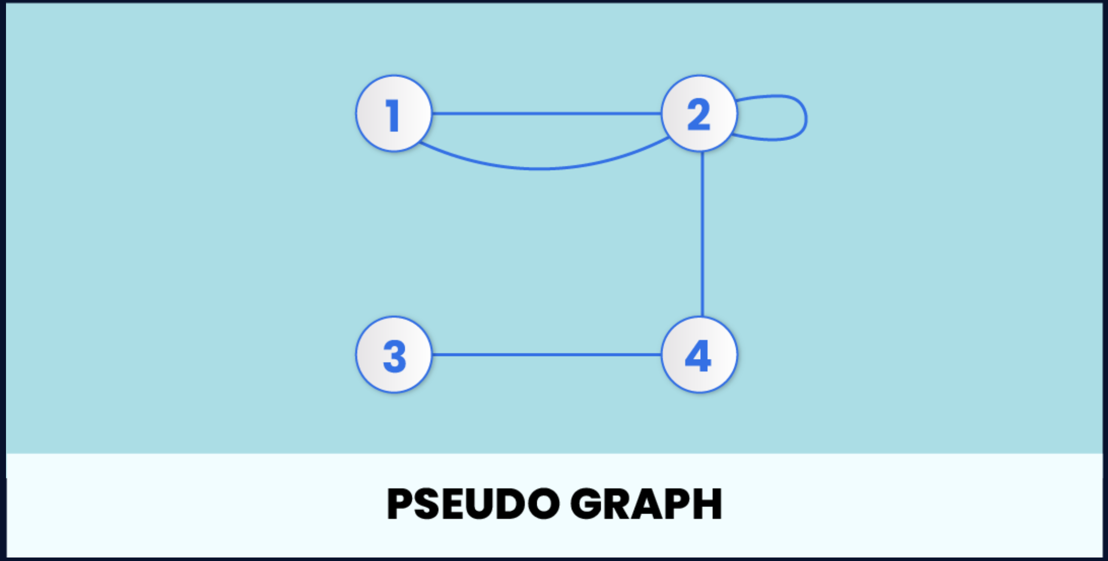
Null graph:

If several vertices but no edge connected them.



Pseudo graph:

If it contains self loop.



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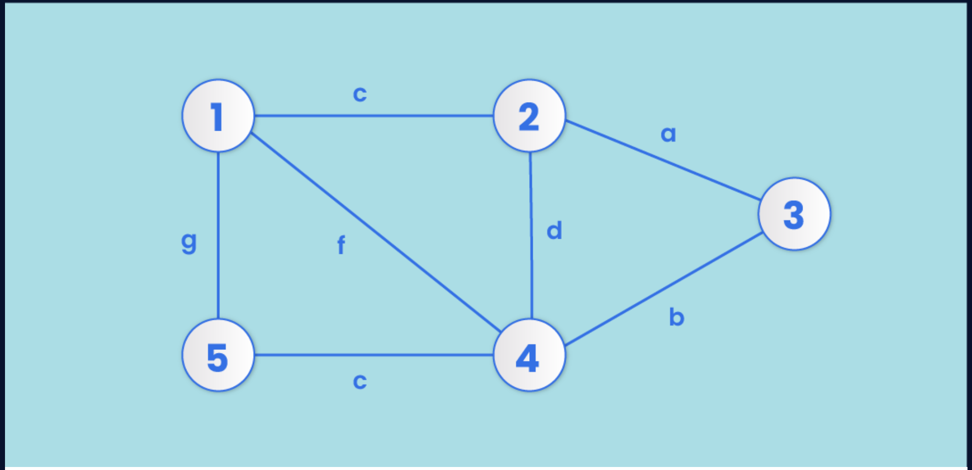
Weighted graph:

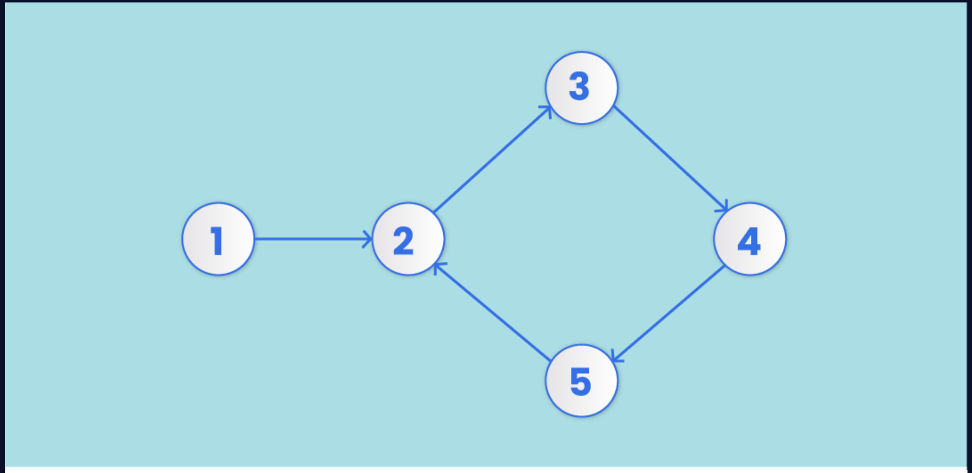
Each edge has a value.



WEIGHTED GRAPH

Directed graph:

Each edge has a direction.



DIRECTED GRAPH

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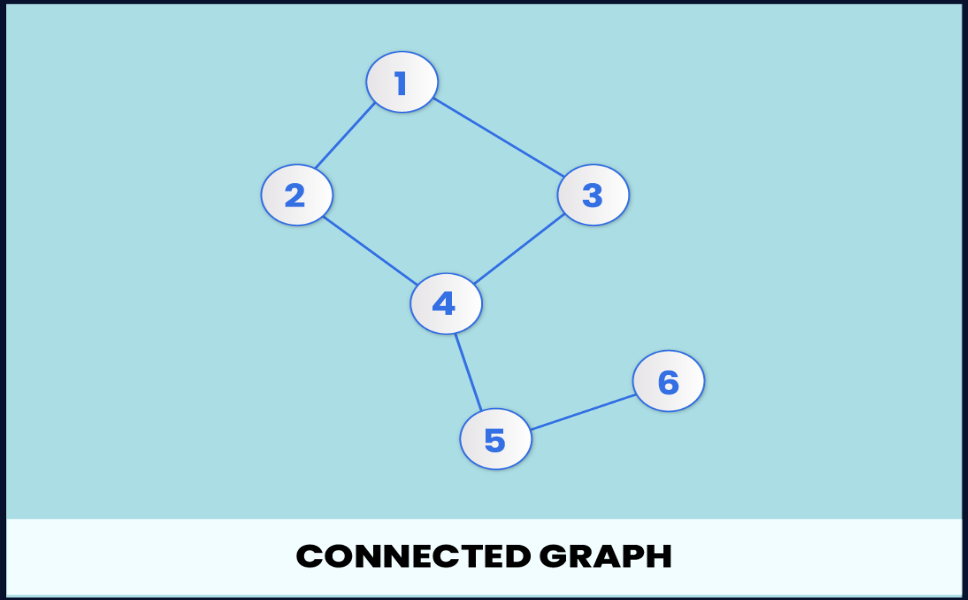
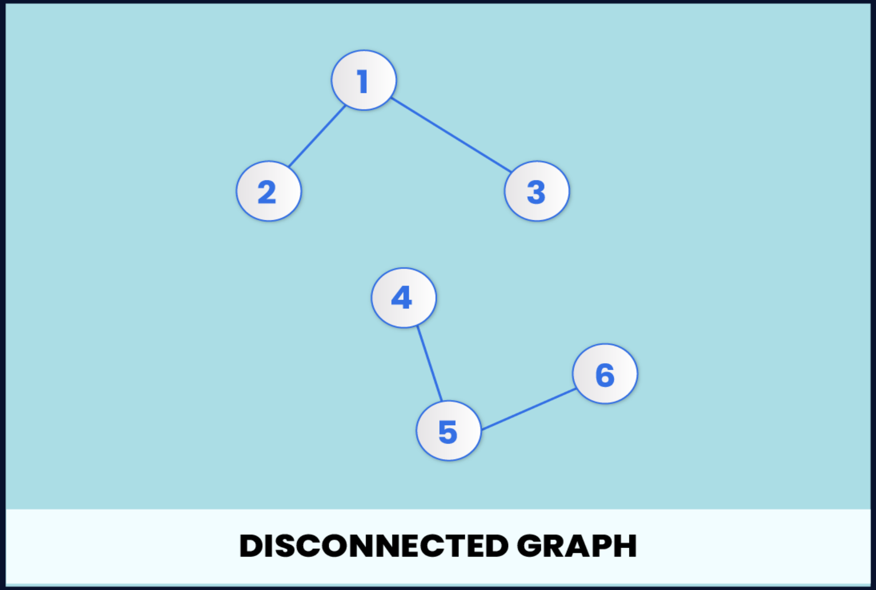
Connected graph:

If there is a path between one vertex and any other vertex.



Disconnected graph:

If there is no edge linking the vertices.



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ADJACENCY MATRIX
&
ADJACENCY LIST

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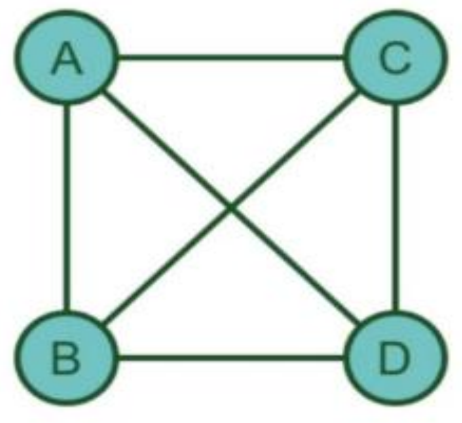
Example

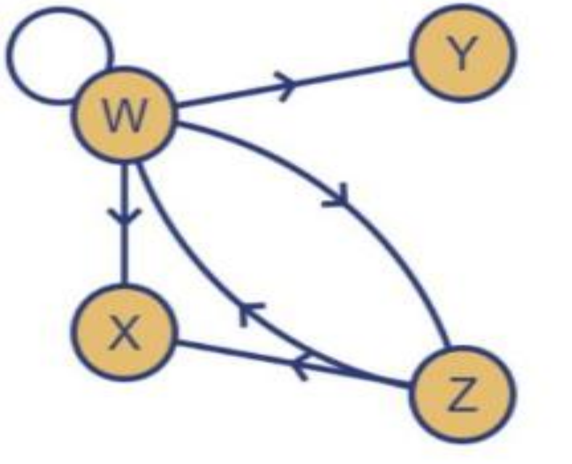


Undirected Graph

Solution

	A	B	C	D
A	0	1	1	1
B	1	0	1	1
C	1	1	0	1
D	1	1	1	0

Example



Directed graph with loop

Solution

	W	X	Y	Z
W	1	1	1	1
X	0	0	0	0
Y	0	0	0	0
Z	1	1	0	0

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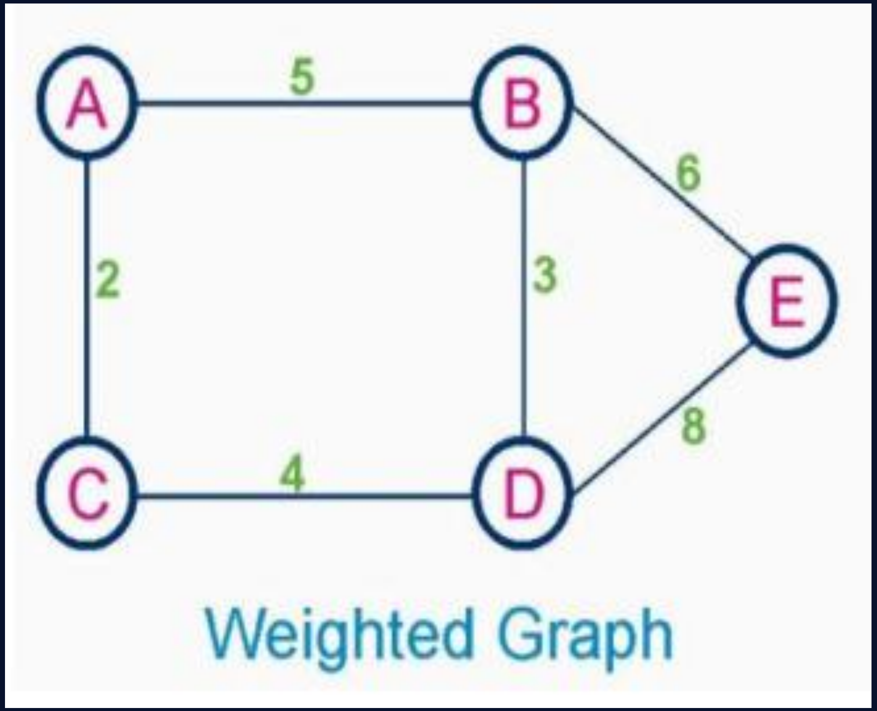
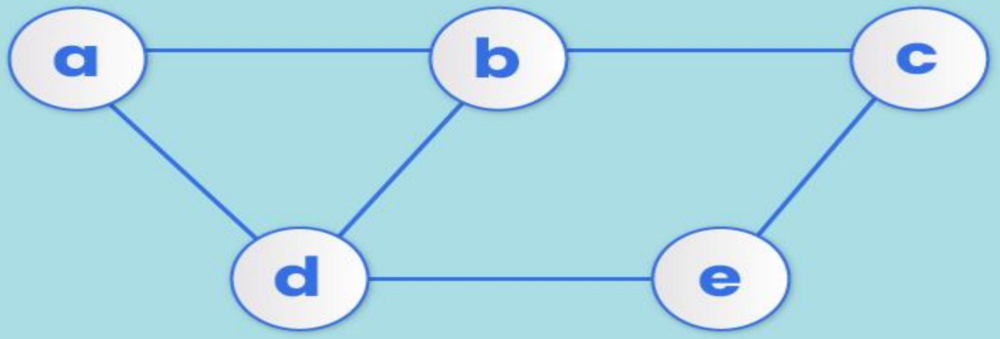
Example



Solution

	A	B	C	D	E
A	0	5	2	0	0
B	5	0	0	3	6
C	2	0	0	4	0
D	0	3	4	0	8
E	0	6	0	8	0

Example



Undirected Graph

Solution

	a	b	c	d	e
a	0	1	0	1	0
b	1	0	1	1	0
c	0	1	0	0	1
d	1	1	0	0	1
e	0	0	1	1	0

Adjacency Matrix

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Example

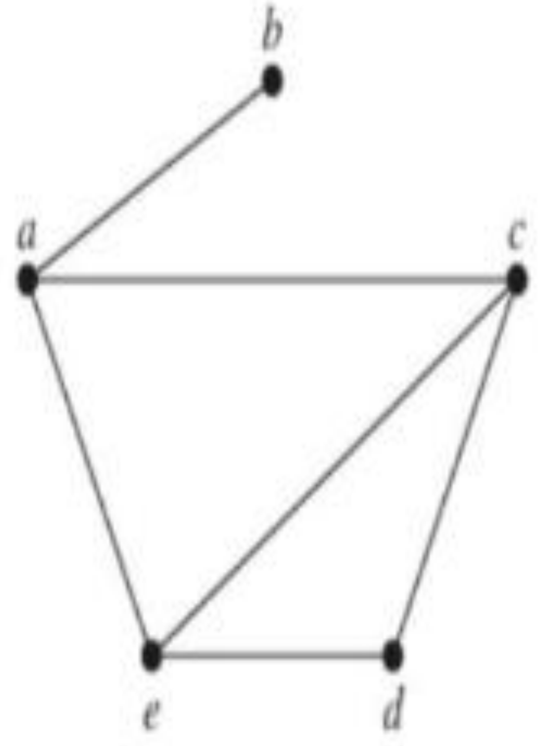

FIGURE 1 A Simple Graph.

Solution

TABLE 1 An Adjacency List for a Simple Graph.

Vertex	*Adjacent Vertices*
a	b, c, e
b	a
c	a, d, e
d	c, e
e	a, c, d

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Example

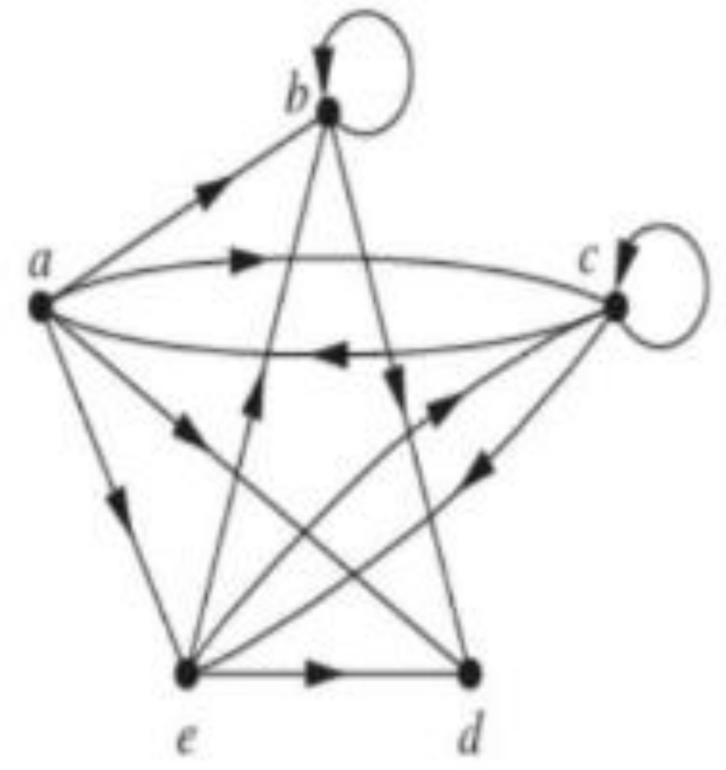

FIGURE 2 A Directed Graph.

Solution

TABLE 2 An Adjacency List for a Directed Graph.

Initial Vertex	Terminal Vertices
a	b, c, d, e
b	b, d
c	a, c, e
d	
e	b, c, d

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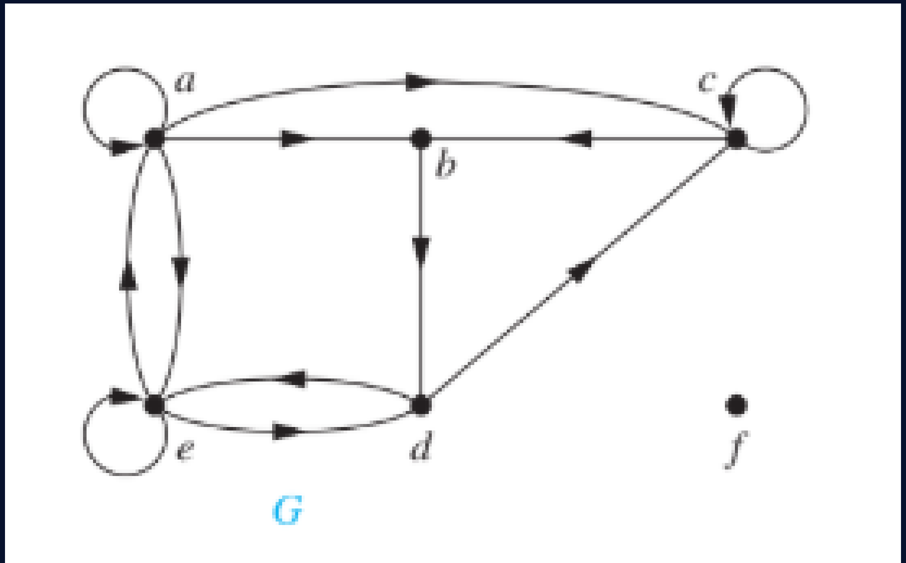
Example

What are the degrees of the vertices is the graphs G ?



Solution

The in-degrees in G are	The out-degrees in G are
deg−(a) = 2 deg−(b) = 2 deg−(c) = 3 deg−(d) = 2 deg−(e) = 3 and deg−(f) = 0	deg+(a) = 4 deg+(b) = 1 deg+(c) = 2 deg+(d) = 2 deg+(e) = 3 deg+(f) = 0

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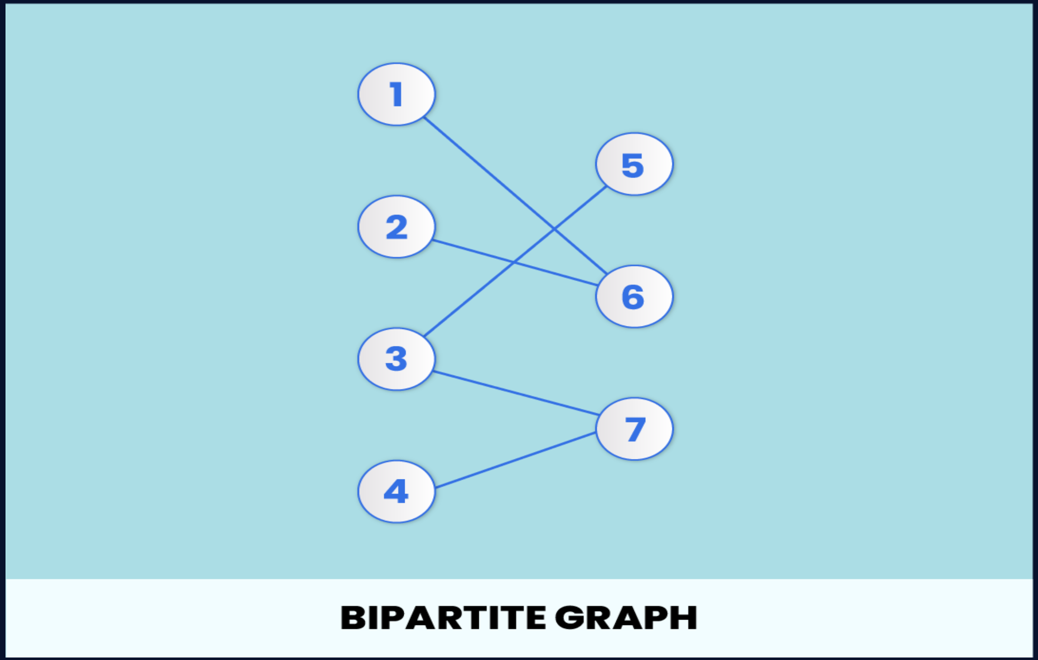
BIPARTITE GRAPH

Definition

A simple graph G is called bipartite if its vertex V can be partitioned into two disjoint sets V1 and V2 such that every edge in the graph connects a vertex in V1 and a vertex in V2.



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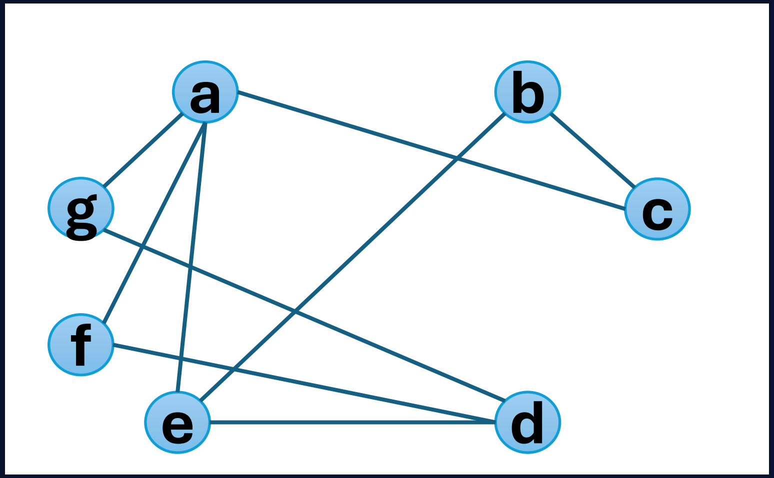
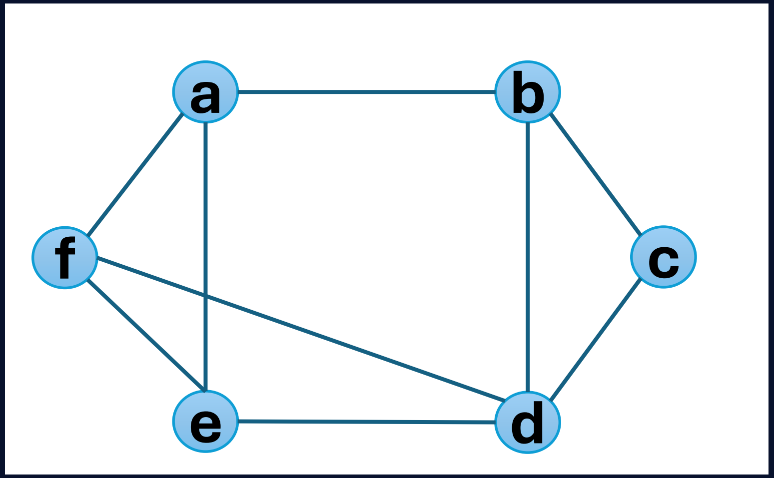
Theorem:

A simple graph is bipartite if and only if it is possible to assign one of two different colors to each vertex of the graph so that no two adjacent vertices are assigned the same color.





Solution

RED	BLUE
a	g, f, e, c
a, b, d	g, f, e, f, c

Solution

RED	BLUE
a	b, f, e

Impossible

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Shortest Path
&
Shortest Distance Problem
Using Dijkstra Algorithm

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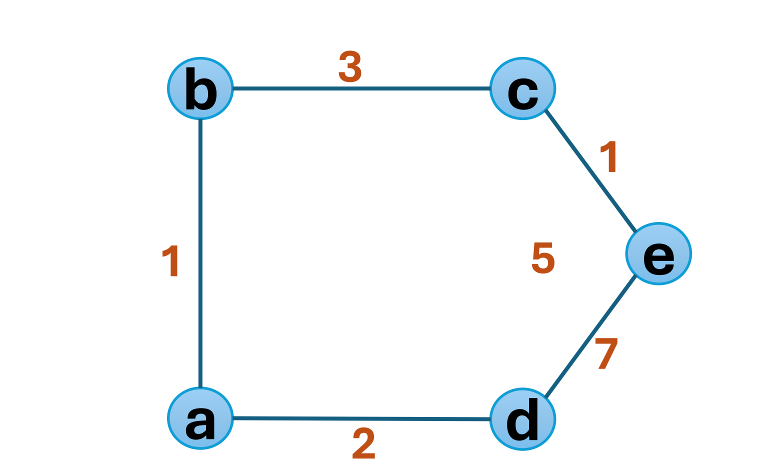
Example

Find the Shortest path and shortest distance from 'a' to 'e' in the given weighted graph, using Dijkstra Algorithm.



Solution

	a	b	c	d	e
a	0	∞	∞	∞	∞
b	0	1	7	2	∞
c	0	1	4	2	∞
d	0	1	4	2	9
e	0	1	4	2	5

Shortest distance: 05

Shortest path:

a → b → c → e

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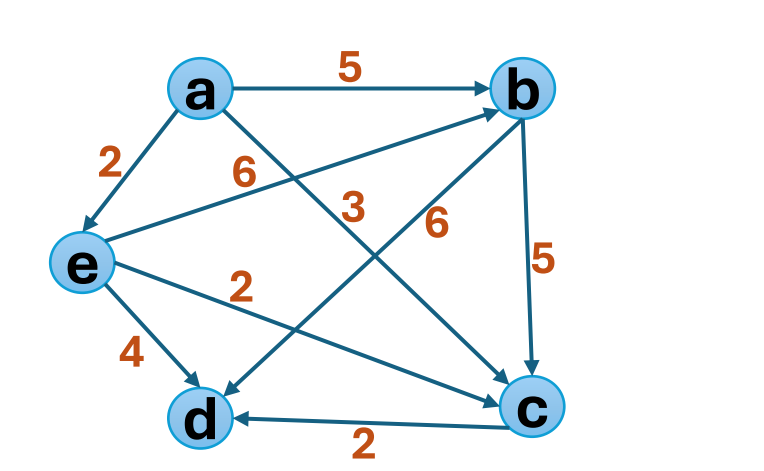
Example

Find the Shortest path and shortest distance from ‘a’ to ‘d’ in the given weighted graph, using Dijkstra Algorithm.



Solution

	a	b	c	d	e
a	0	∞	∞	∞	∞
b	0	5	3	∞	2
c	0	5	3	6	2
d	0	4	3	5	2
e	0	4	3	5	2

Shortest distance: ?

Shortest path: ?

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Tree

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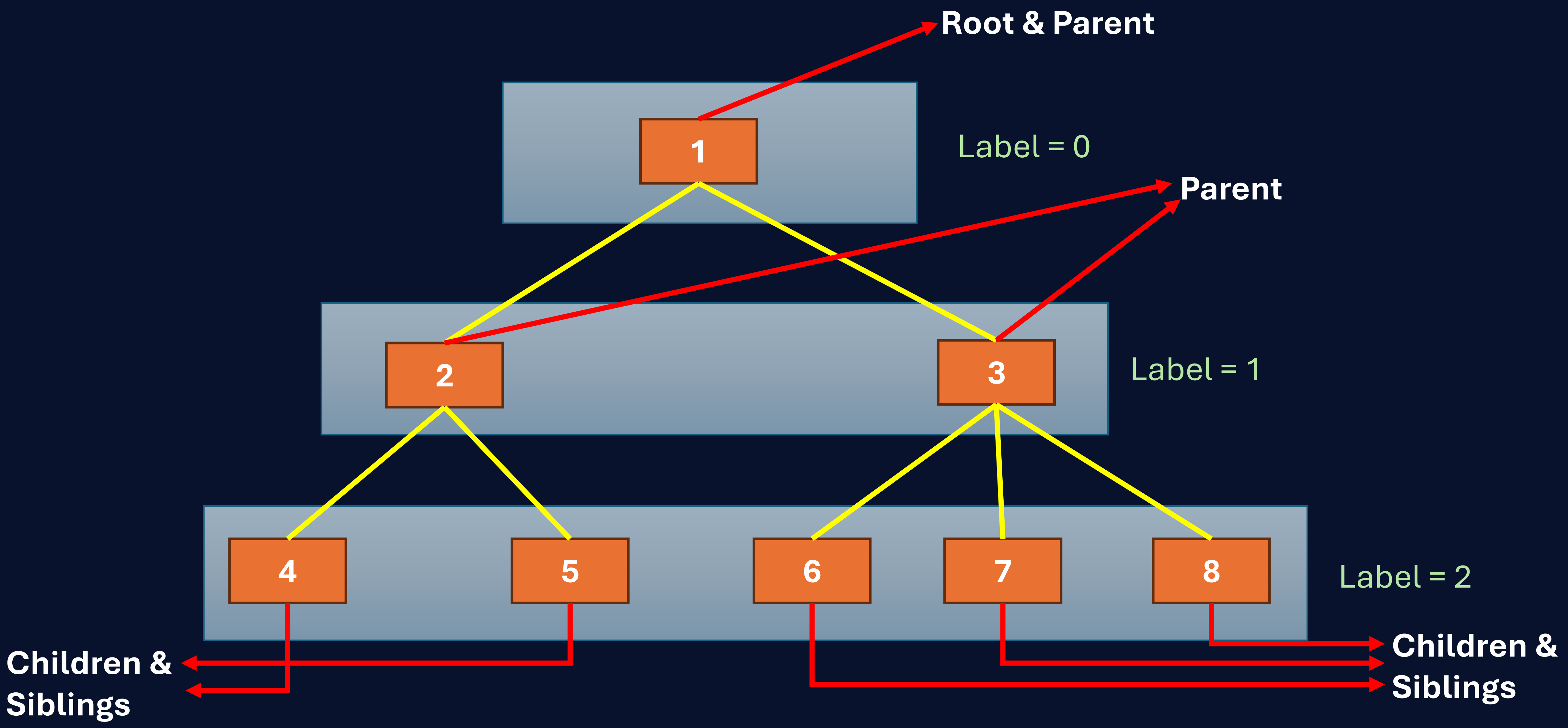

Root, Parent, Children, Siblings, Internal Vertices
Leaves, Label, Height

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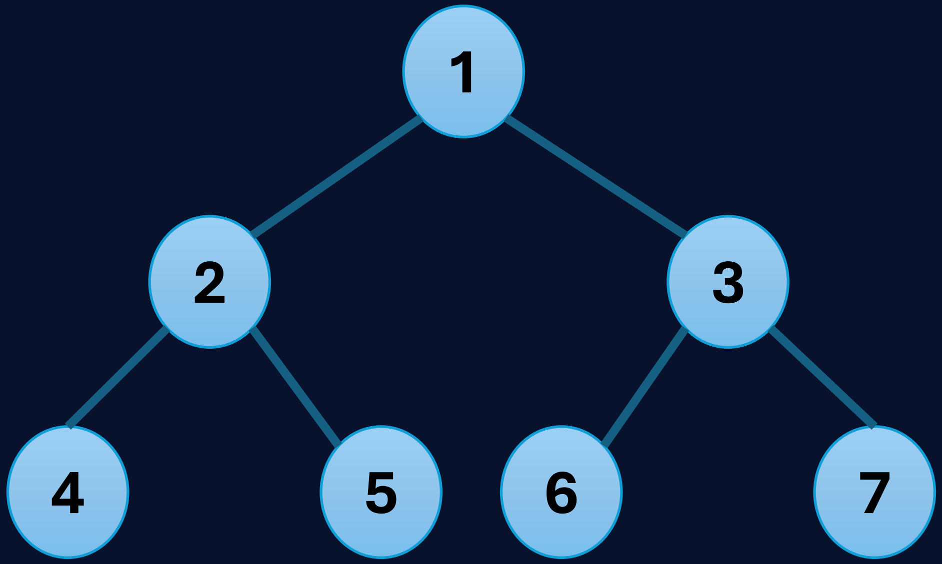
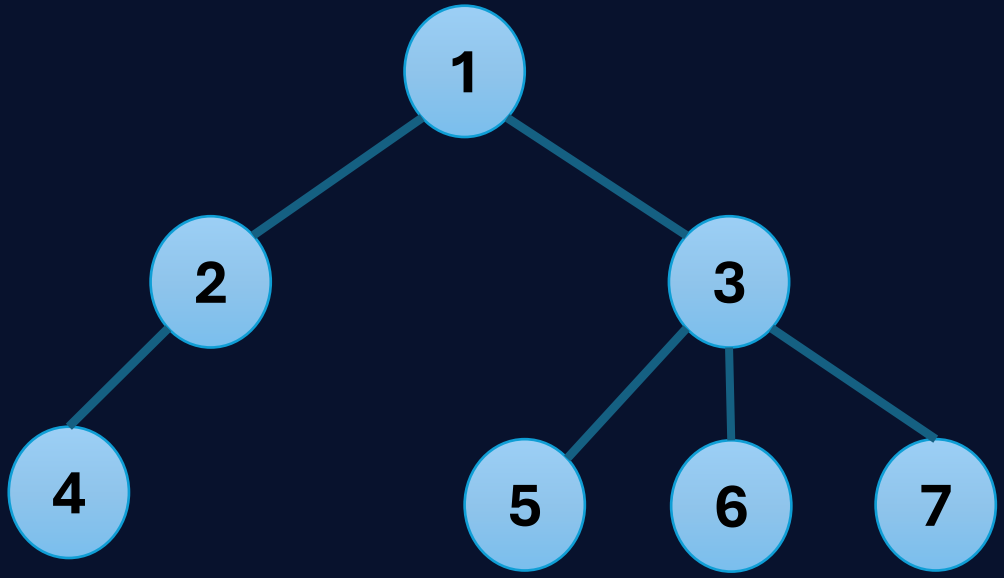
1. M-ary tree
2. Full m- ary tree



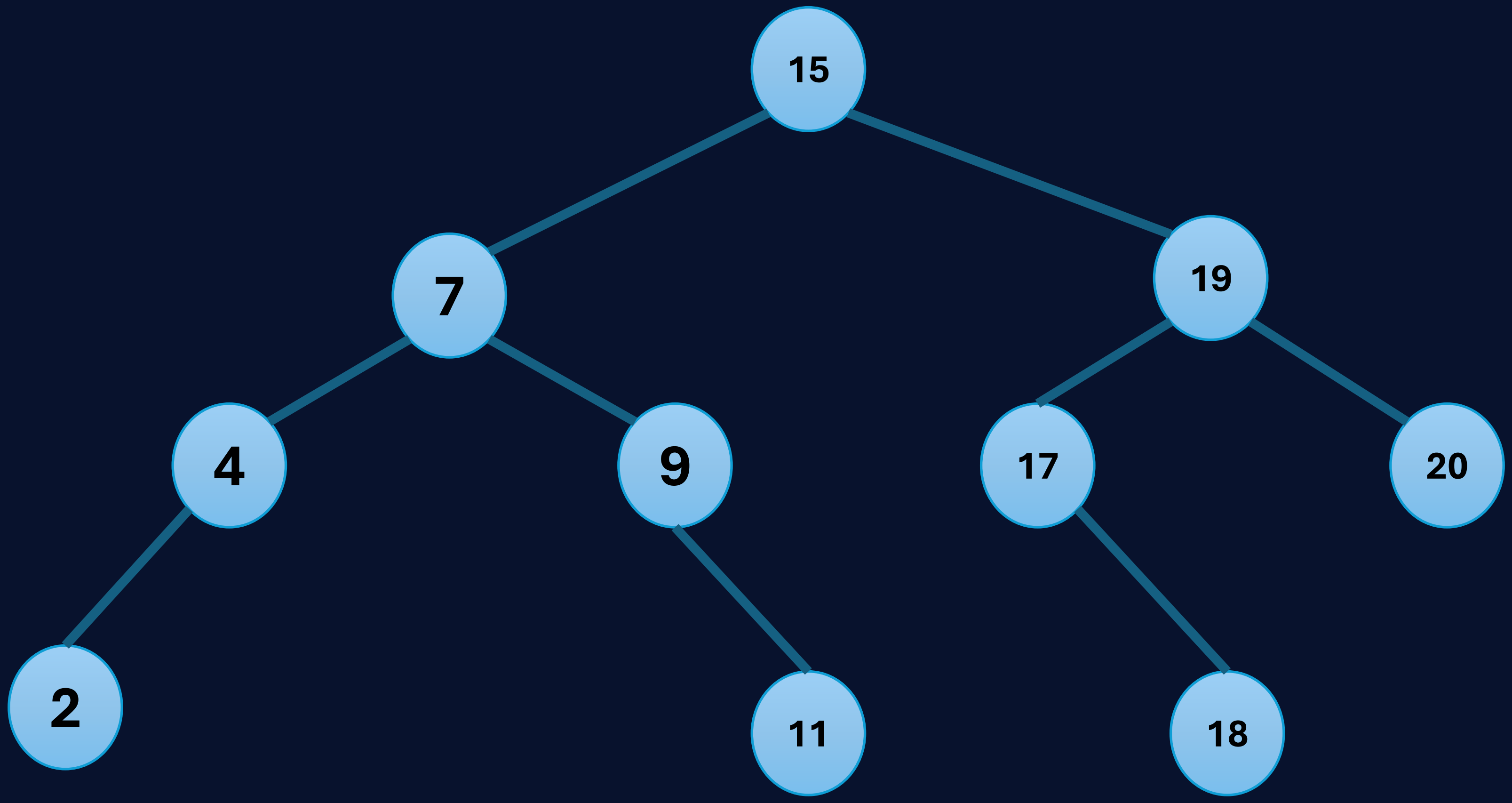
Full 2-ary tree



3-ary tree

Binary Search Tree

Create a binary search tree using the tallowing data:
15, 7, 19, 9, 4, 2, 17, 11, 20, 18



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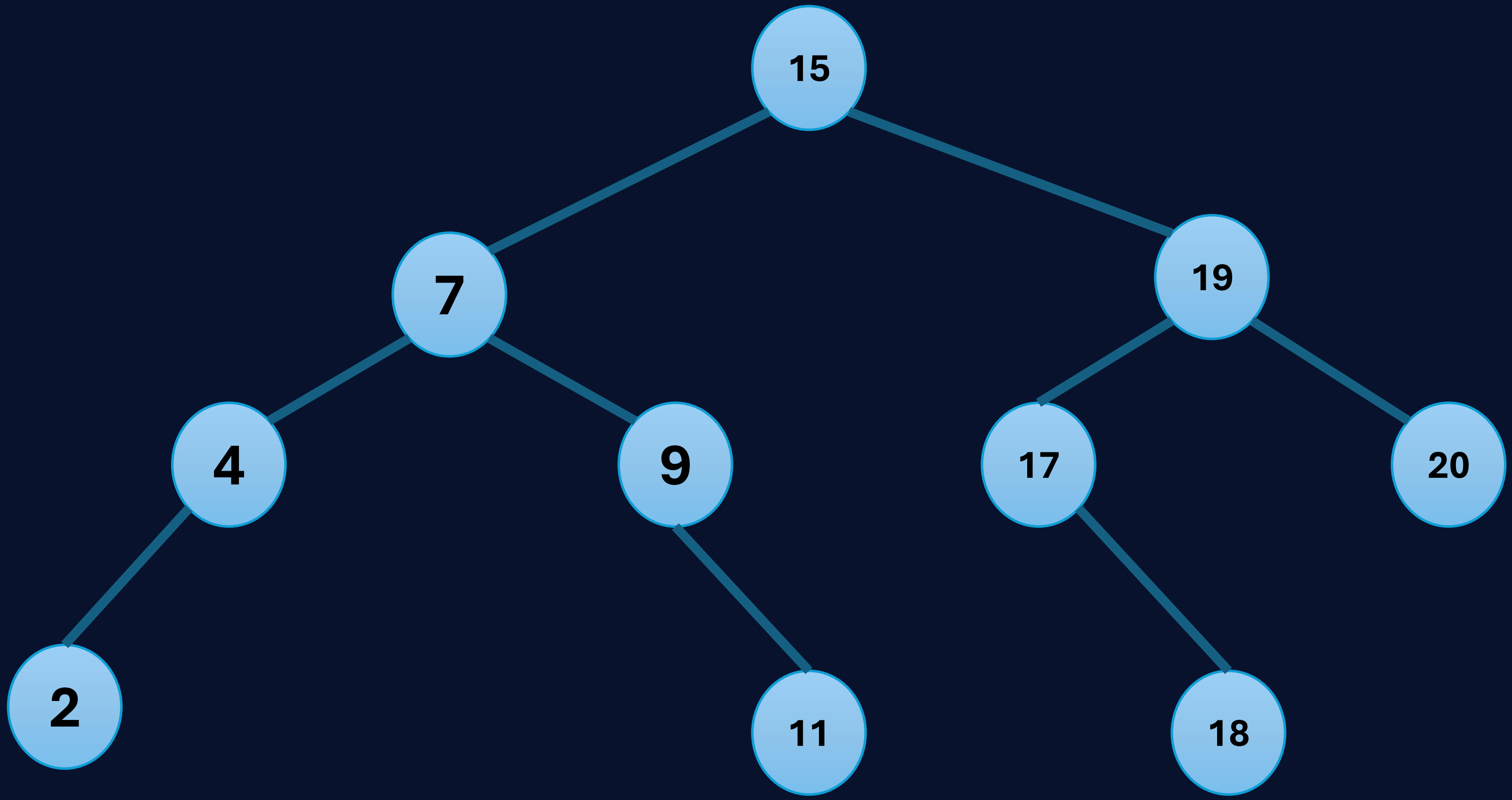
Tree Traversal

1) **Pre – order: Root, Left, Right.**
2) **In – order: Left, Root, Right.**
3) **Post – order: Left, Right, Root.**

Example:



Solution:

1. **Pre – order:**
15, 7, 4, 2, 9, 11, 19, 17, 18, 20

2. **Post – order:**
2, 4, 11, 9, 7, 18, 17, 20, 19, 15

3. **In – order:**
2, 4, 7, 9, 11, 15, 17, 18, 19, 20

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Number Theory

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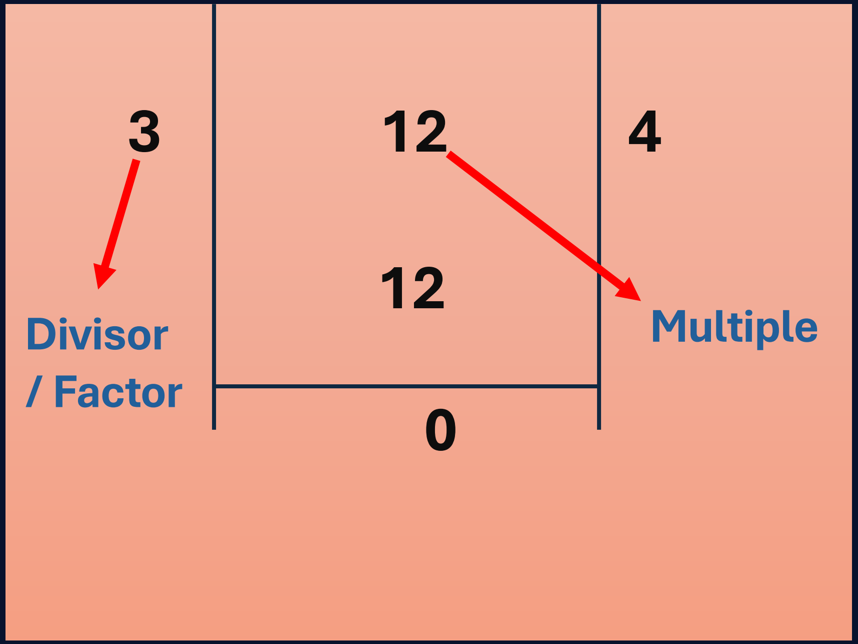
Division:

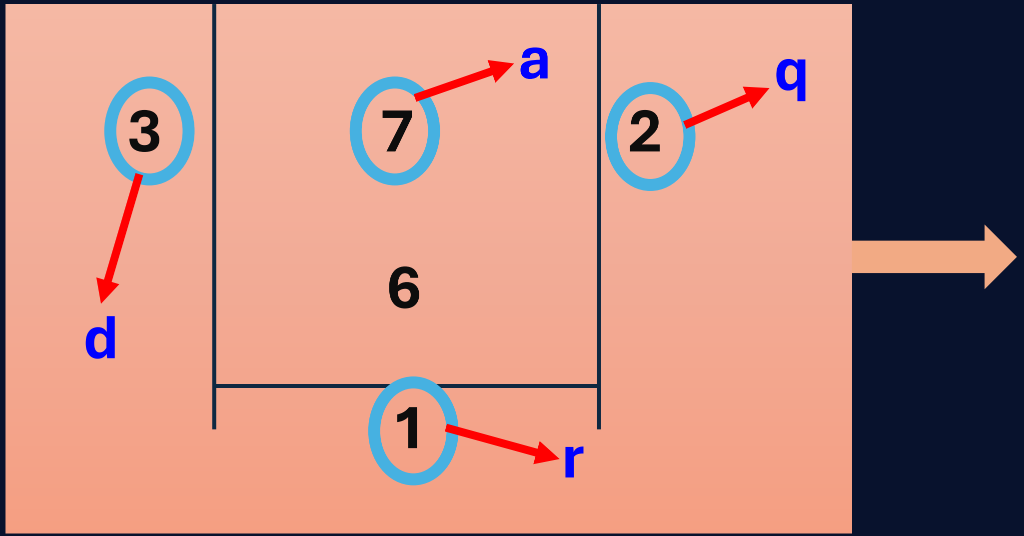
If a and b are integers with $a \neq 0$, we say that a divides b if there is an integers c such that $b = ac$, or equivalently if b/a is an integer.

- ✓ Real number $\mathbb{R} = (-\infty, \infty)$
- ✓ Natural number $\mathbb{N} = \{1,2,3,4,5,6....\}$
- ✓ Integer number = $\{.....-2,-1,0,1,2.....\}$
- ✓ Rational = $\{p/q,\ p.q \in \text{integer},\ q \neq 0\}$



The Division Algorithm:

Let a be an integer and b a positive integer. Then there are positive integers q and r, with $0 \le r < d$, such that $a = dq + r$.



d = divisor
a = multiple
q = quotient
r = reminder

$7 = 3 \times 2 + 1$

$a = d \times q + r$

Example 01:

What are the quotient and reminder when 101 is divide by 11 ?

Solution: →

11	101	9
	99	
	2	

→ q = 9 = 101 div 11
r = 2 = 101 mod 11

Example 02:

What are the quotient and reminder when -11 is divide by 3 ?

Solution: →

3	-11	-4
	-12	
	1	

→ $0 \leq r < 3$

Primes

An integer P is greater than 1 is called prime if the only positive factor of P are 1 and P. A positive integer that is greater than 1 and is not prime is called composite.

2,3,5,7,11,13.........

Example

100 = 2 x 2 x 5 x 5

641 = 641

999 = 3 x 3 x 3 x 37

1024 = 2 x 2 x 2 x 2 x 2 x 2 x 2 x 2 x 2 x 2 = 2^{10}

Find the prime factorization of 7007 ?

Solution: 7007 = 7 x 1001

= 7 x 7 x 143

= 7 x 7 x 11 x 13

Note: Divide the number n → $\sqrt{n}$

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- ➢ **Greatest Common Divisor (GCD)**
- ➢ **Least Common Multiple (LCM)**

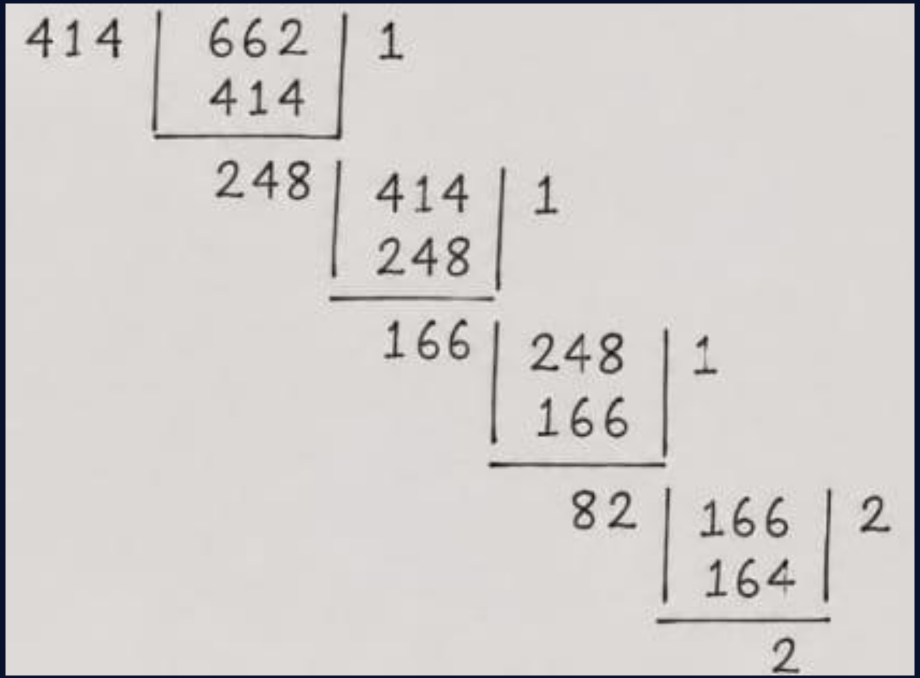
Example

24 = 2,3,4,8,12,6
36 = 2,3,4,6,9,12,18

2,3,4,6,12

Find the GCD of 414 and 662 ?

Solution:



GCD = 2

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The Euclidean Algorithm

Let $a = bq + r$, where a, b, q and r are integers. Then GCD (a, b) = (b, r).

Example

a b q r

$a = bq + r$

662 = 414 x 1 + 248
414 = 248 x 1 + 166
248 = 166 x 1 + 82
166 = 82 x 2 + 2
82 = 2 x 41 + 0

GCD (662, 414) = GCD (414, 248) =GCD (166, 82) =GCD (82, 2) = 2

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Cryptography

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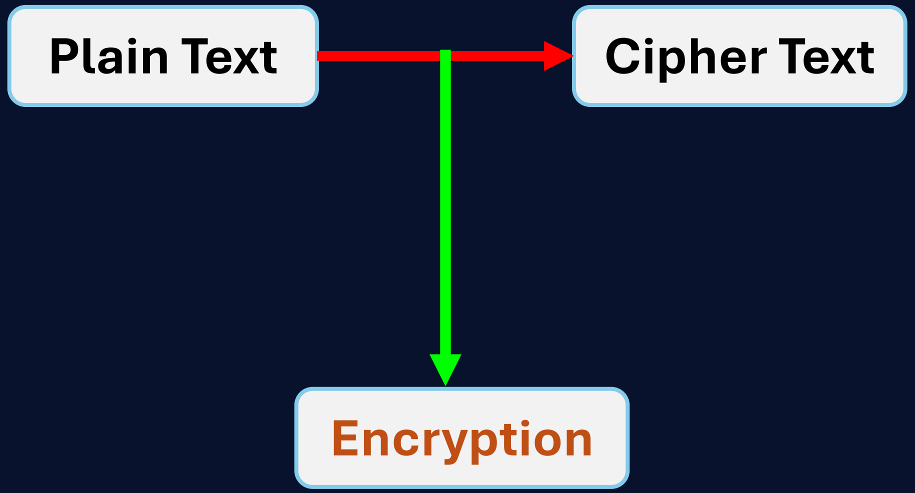
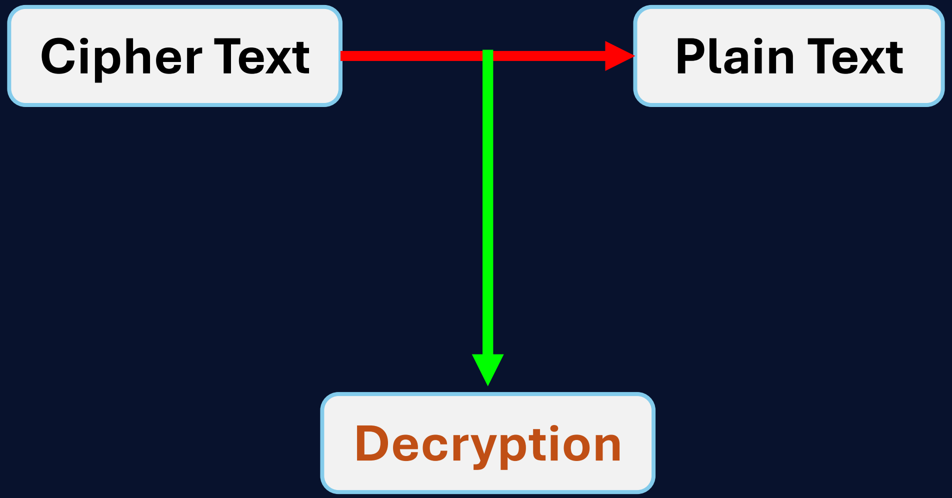
Definition

Cryptography is a method or technique by which ordinary information or message is converted into a secret code so that no one other than the intended recipient can read or understand it.





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Key = 3 caser cryptography

Classical Shift Cryptography

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Example

C S E = Plain Text

F V H → **Encryption**

C S E → **Decryption**

Sets

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Definition

A set is an unordered collection of objects, called elements or members of the sets.

Example

Roster method

I. The set v of all vowels in the English alphabet can be expressed as V = {a, e, i, o, u}

II. The set of positive integers less than 100 can be denoted by {1,2,3.....,97,98,99}

Set builder method

III. The det O of all positive integers less than 10 can be written as O = {x | x id an odd positive integer less than 10 }

Or, $O = \{ x \in \mathbb{Z}^{+} \mid x$ is odd and $x<10$

Some important sets

$\mathbb{N}$ = {0,1,2,.....}, the set of natural numbers.

$\mathbb{Z}$ = {.....-2,-1,0,1,2,.....}, the set of integers.

$\mathbb{Z}^+$ = {1,2,3,.....}, the set of positive integers.

Q = {p/q | p $\in \mathbb{Z}$, q $\in \mathbb{Z}$, and q $\neq$0}, the set of rational numbers.

$\mathbb{R}$ = The set of real numbers.

$\mathbb{R}^+$ = The set of positive real numbers.

C = The set of complex numbers.

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Subsets

The set A is a subset of B if and only if every element of A is also an element of B. W e use the rotation A ⊆B to indicate that A is a subset of the set B.

The size of a set

Let S be a set. If there are exactly n distinct elements in S and x is a non-negative integer we say that S is a finite set and n is the cardinality of S. Cardinality of S denoted by |S|.

Example

Let S be the set of all letters in the English alphabet. Then |S| = 26.

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The size of a set

The power set of a set S is the set of all subsets of the set S. The power set of S is denoted by P (S).

Example

1. What is the power set of the set {0,1,2} ?

→ P {0,1,2} = { Ø, {0}, {1}, {2}, {0,1}, {0,2}, {1,2}, {0,1,2} }

2. What is the power set of the empty set ?

→ P(Ø) = {Ø}

3. What is the power set of the set {Ø} ?

→ P{Ø} = { Ø, {Ø} }

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Cartesian Product

Let A and B be sets. The cartesian product of A and B, denoted by A x B, is the set of all ordered pairs (a,b), where a ∈ A and b ∈ B. Hence

$A \times B = \{ (a,b) \mid a \in A \wedge b \in B \}$

Example

What is the Cartesian product of A = {1,2} and B = {a,b,c} ?

→ The Cartesian product A x B is

$A \times B = \{ (1,a), (1,b), (1,c), (2,a), (2,b), (2,c) \}$

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Set operations

Union

Let A and B be sets. The union of the sets A and B, denoted by A ∪ B, is the set that contains those elements that are either in A or in B, or in both.

$$A \cup B = \{ x \mid x \in A \vee x \in B \}$$

Example

The union of the sets {1,3,5} and {1,2,3} is the set {1,2,3,5} ; that is , {1,3,5} ∪ {1,2,3} = {1,2,3,5}.

Intersection

Let a and B be sets. The intersection of the sets A and B, denoted by A ∩ B, is the set containing those elements in both A and B.

$$A \cap B = \{ x \mid x \in A \wedge x \in B \}$$

Example

The intersection of the set {1,3,5} and {1,2,3} is the set {1,3}; that is, {1,3,5} ∩ {1,2,3} = {1,3}.

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Difference

The difference of A and B, denoted by A - B, is the set containing those elements that are in A but not in B.

$$\mathbf{A - B = \{ x \mid x \in A \wedge x \notin B \}}$$

Example

The difference of {1,3,5} and {1,2,3} is the set {5}. That is, {1,3,5} – {1,2,3} = {5}

Complement

Let U be the universal set. The complement of the set A, denoted by Ā, is the complement of the set A is U – A.

$$\mathbf{\bar{A} = \{ x \in U \mid x \notin A \}}$$

Example

Let A be the set of positive integers grater than 10 (with universal set the set of all positive integers).
Then Ā = {1,2,3,4,5,6,7,8,9,10}.

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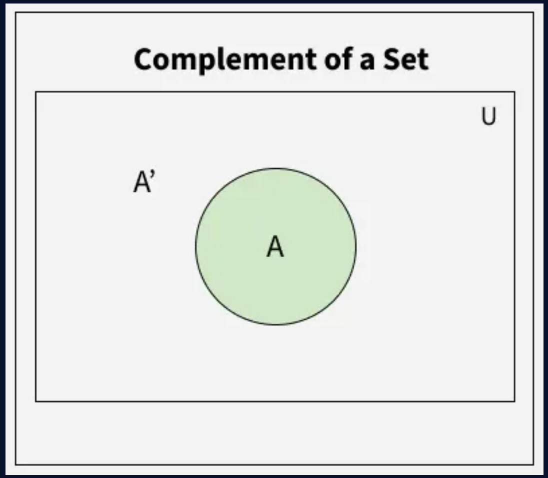
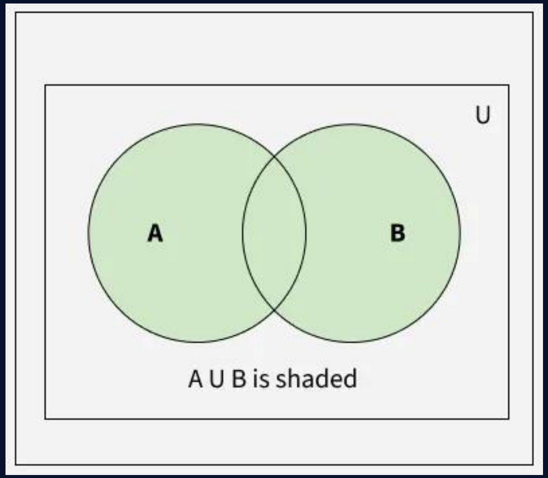
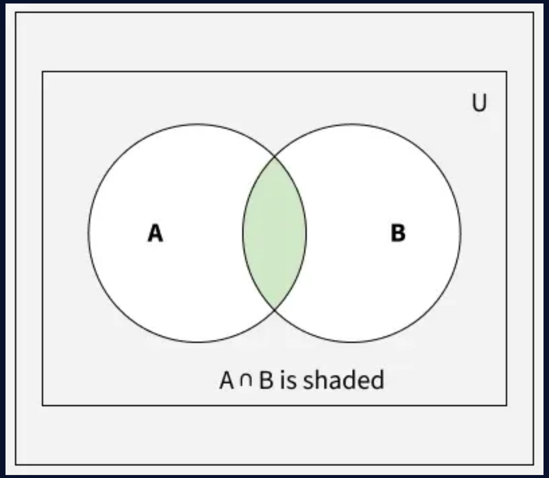
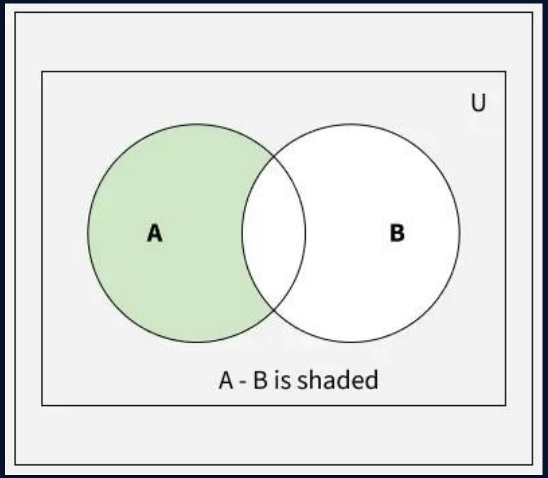
Diagram of set









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Function

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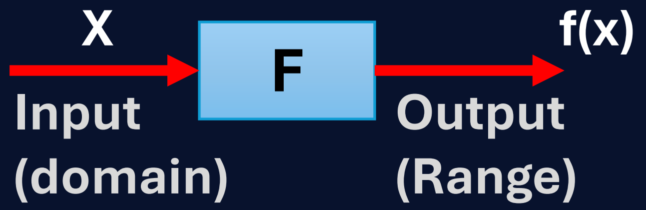
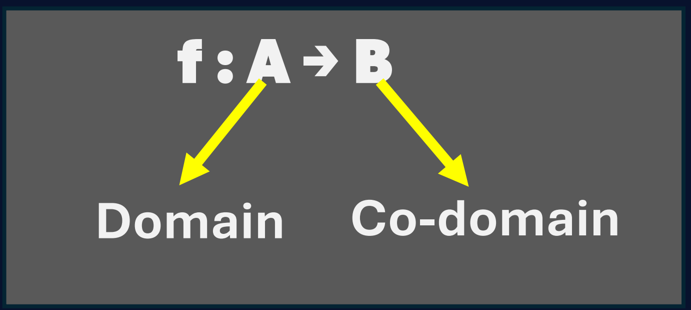
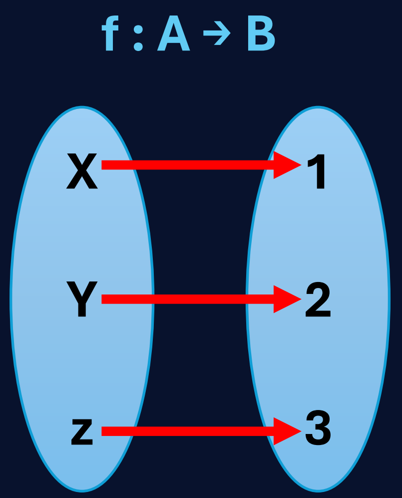
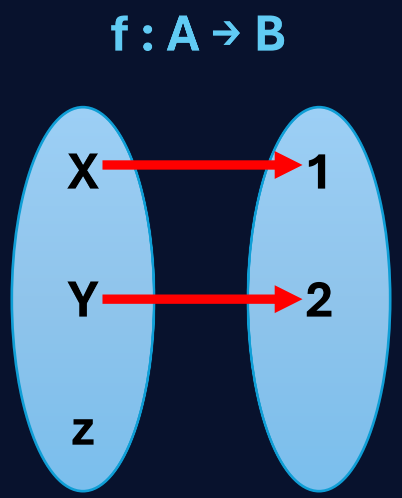




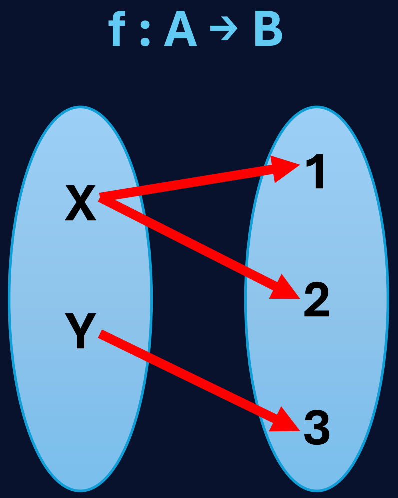
- A function is a special type of relation, where
  → Every element in the domain is included.
  → Any input produce only one output.



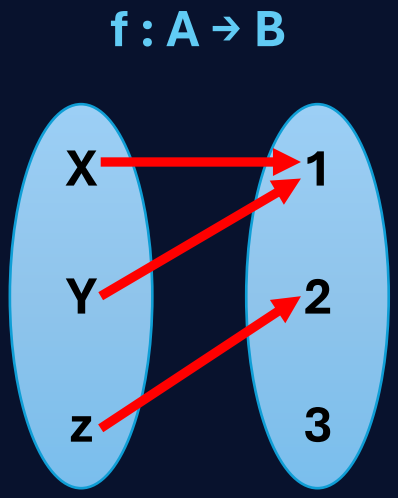
A Function



Not a Function



Not a Function



A Function

- A function f take elements from a set (domain) and relates to element in a set (the co-domain)
  → All outputs together are called range.

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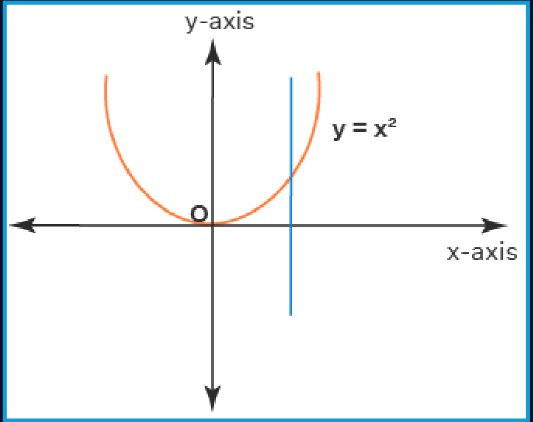
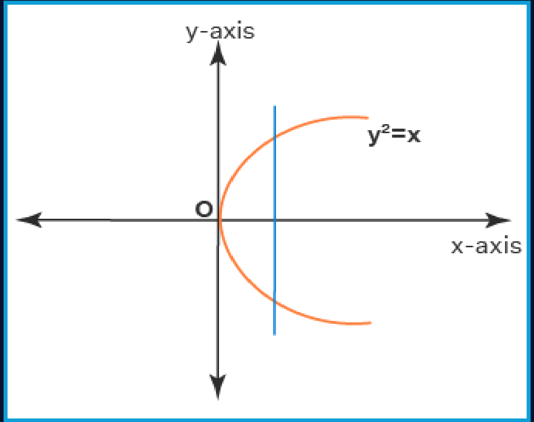
Vertical lint test

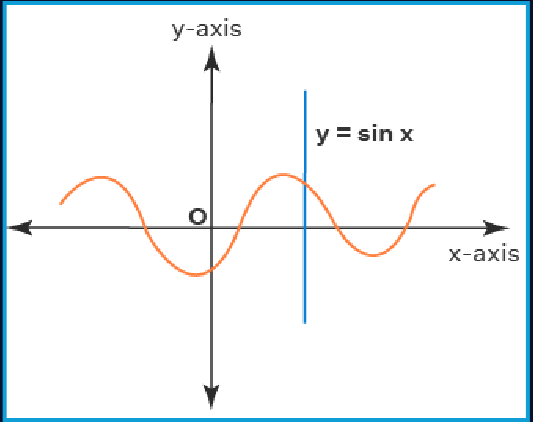
If all vertical lines interact a curve at most once then the curve represents a function.



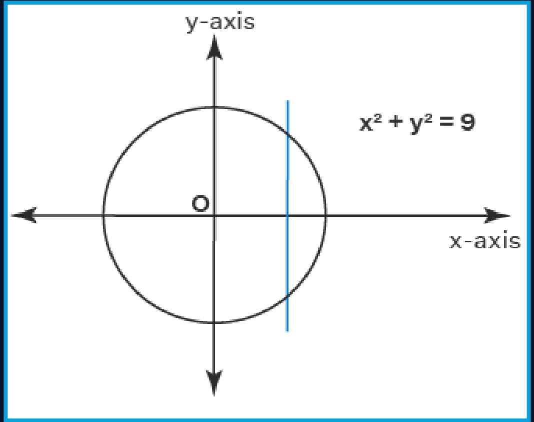
A function



Not a function



A function



Not a function

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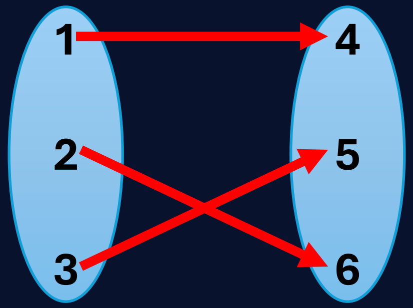
One-to-one Function (Injective)

A function f is one to one if every element of the range of f corresponds to exactly one element of the domain of f.

f : A → B

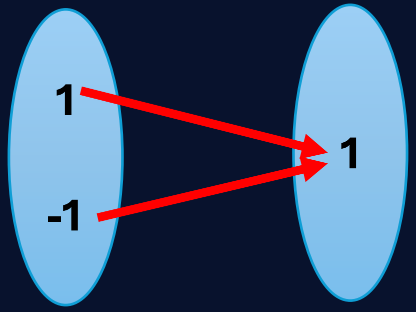


Example

Determine whether the function f(x) = x^2 from the set integer to the set of integers is one-to-one.

Solution:

The function f(x) = x^2 is not one-to-one. Because, if x = 1, f(x) = x^2 = 1 again if x = -1, f(x) = 1



Function but not one-to-one

Example

Determine whether the function f(x) = x+1 from the set of real numbers to itself is one-to-one.

Solution:

f(x) = x+1 is one-to-one function. To demonstrate this,
If f(a) = f(b); for all a,b ∈ domain of f
→ a+1 = b+1
→ a=b

Onto Function (Surjective)

Co-domain = Range

Example

If the function f(x) = x^2 from the set of integers to the det of integers onto ?

Solution:

$$\left.\begin{array}{l}\text{Co-domain} = \mathbb{Z} = \text{the set of integers} \\ \text{Range} = [\ 0,\infty\ )\end{array}\right\}\ \text{Not onto}$$

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Example

Is the function f(x) = x+1 from the set of integers to the set of integers onto ?

Solution:

$$\left.\begin{array}{l}\text{Co-domain} = \mathbb{Z} = \text{the set of integers} \\ \text{Range} = \mathbb{Z} = \text{the set of integers}\end{array}\right\} \text{Onto function}$$

Bijective Function

If f is both one-to-one and onto.
Injective + Surjective

Inverse Function

A function is invertible if it is bijective.

f: A → B
f^{-1} **: B → A**

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Example

Let f: $\mathbb{Z} \rightarrow \mathbb{Z}$ be such that f(x) = x+1. If F invertible and if it is, that is its inverse ?

Solution:

The function f has an inverse because it is a bijective function (from previous example). To inverse this

f(x) = x+1
Or, y = x+1
Or, x = y-1
Or, f^{-1} (y) = y -1
Or, f^{-1} (x) = x -1

y = f(x)
x = f^{-1} (y)

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